

Comparison of mayfly (Ephemeroptera) taxocenes of permanent and intermittent Central European small streams via species traits

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Abstract: The recurrent drying out of small streams in past decades has shown an urgent need to pay attention to the impact of global climate change. The objectives of this study were to describe the effect of drying out on the composition of the mayfly taxocene and evaluate the relevance of individual species traits for survival of mayflies to drying out. The mayfly taxocenes of two model localities, one at an intermittent and one at a permanent brook, were investigated in 2002, 2003 and 2005. Compared with the permanent stream, the taxocene of the intermittent stream was short of nine species, foremost rheobionts and high oxygen demand species. To explain further differences between both stream types in survival and recolonisation ability, 15 species traits were evaluated. These included so-called “ecological traits” (e.g., habitat and substrate range, density, distribution, current velocity adaptation) and “biological traits” connected with life cycle and larval/adult adaptations. Species showing the highest number of advantageous traits (with only exception of *Electrogena* sp. cf. *ujhelyii* – species of taxonomically unclear status) were able to successfully survive under the unfavourable conditions of the intermittent brook. Biological traits considered more important in many respects seem to be good predictors for assessing sensitivity to extreme temperature changes, hydrological regime fluctuations and the survival/recolonisation ability of species in exposed habitats.

Key words: Ephemeroptera; drought survival; species traits; life cycles; intermittent stream; Czech Republic

Introduction

The climate has globally changed many times over the course of the Earth’s history. At present, we are in a warm interglacial period, but over the past two million years there have been numerous glacials. The last one (Würm or Vistula Glaciation) had a greater impact on biota because this period exerted a crucial effect on the selection and composition of contemporary faunas (and biota in general), the phylogeny and biodiversity of most animal groups and numerous processes at population or biosystem levels. Geological records show that past climate changes have not always been gradual; rather there have been numerous rapid changes, sometimes over period as short as centuries or even decades (Bonada et al. 2007b), with principal changes able to be traced even within modern human’s history. For instance, starting around 1550 and lasting until around 1850 was a period of cold temperature called the “Little Ice Age” (Lamb 1995).

There are several future climate scenarios for individual regions of the world. Taking into consideration Europe, there are several alternatives, which identically expect local and/or regional temperature increases and more variable precipitation patterns (Benis-

ton et al. 2007; Tapiador & Sánchez 2008). In Central Europe, no substantial changes of annual precipitation are anticipated unlike seasonal precipitation distribution (Pišoft et al. 2004; Kyselý & Beranová 2009). Winter–spring precipitation is expected to increase and summer–autumn precipitation decrease (Bonada et al. 2007b). Temperatures in Europe warmed by 0.5°C between the mid-1800s and 1940 but cooled again over the next 25 years. Since then temperatures have been rising and current predictions forecast increases of annual mean air temperatures by as much as 8°C by the end of century in some regions (cf. e.g., IPCC 2001).

Climate change impacts both terrestrial and aquatic parts of the ecosystems. Considerable attention is paid to fluvial systems, which can be affected by changes of hydrologic regimes, primarily in terms of extreme hydrological events such as high discharge and drying out. Over the past decade in Central Europe, besides the phenomenon of summer drought (an unusual event in this region), there have also been rising occurrences of summer floods. For instance, extreme floods occurred in the Czech part of the Danube basin in July 1997 and in the Elbe basin in August 2002. By contrast, the summer in 2003 was extremely hot and dry in Central Europe.

The lack or surplus of water can result in changes in the taxonomical composition and functional structure of assemblages (Bonada et al. 2007a), which could have far-reaching impacts on the self-cleaning processes of water (and consequently on oxygen regime), as well as on the productivity of freshwater biotopes. The impact of extremely low discharges and the drying out of various freshwater habitats was a frequent topic of research in regions with common occurrences of these events, for example, Australia (e.g., Brock et al. 2003; McMahon & Finlayson 2003; see also a review by Boulton 2003) and South and North America (Miller & Golladay 1996; del Rosario & Resh 2000; Smith et al. 2001; Covich et al. 2003). In Europe, the drying out of streams has especially been studied in the south of the continent (e.g., Pires et al. 2000), focusing on the differences in the taxonomical and functional structure of benthic assemblages between streams of temperate region and the Mediterranean, including temporary streams (Bonada et al. 2007a). The Czech Republic, which is situated just in the centre of the European continent and at the boundaries of three sea drainage areas (the North Sea, Baltic Sea and Black Sea), shows a high percentage of small (first to fourth order by Strahler) running waters, which represent approximately 90% of the total length of Czech rivers and streams. These types of streams are vulnerable to anomalous discharges. Small perennial streams, as well as intermittent streams natural in origin can be found in several regions of the Czech Republic. These regions are defined by specific geological conditions (karstic, e.g., the Moravian Karst and Cretaceous regions, e.g., parts of the Czech Plateau). Both perennial and intermittent streams also exist in heavily altered areas (deforested or ameliorated areas, e.g., the Pannonian lowlands in the southeast of the Czech Republic). These regions were most affected by extreme drought and above average summer temperatures in 2003.

In this extremely dry period (2002–2003), studies of macroinvertebrate assemblages of two nearby brooks on the boundary of the Central European highlands and Pannonian lowlands (Illies 1967) were carried out. One of these brooks dried out. The results of the research conducted in the intermittent brook (the Gránický brook) were published (Řezníčková et al. 2007a), whereas the data on the second permanent brook (the Klaperův brook) were left unpublished in a diploma thesis (Nyklová 2006). These unpublished data from the permanent brook, which showed similar abiotic characteristics to the Gránický brook, were used to define the reference state in this study. Comparison of these results stimulated follow-up research, which took place in 2005 in the intermittent brook (the Gránický brook). The results are being processed (Řezníčková et al. 2007b). In comparison with the permanent brook, the intermittent brook showed evident and stable differences in (among others) the composition and species richness of mayfly taxocenes, which were not interpretable by the impacts of pollution or hydromorphologic dissimilarities of both brooks, which

were negligible. Thus, this drying out seemed to be the only disturbance. Surviving a dry period heavily depends on the characteristics of the species and their biological and ecological traits. The species traits important for overcoming this type of disturbance have not yet been summarised, and there is also a lack of detailed information on particular species. Therefore, the objectives of this study were to (i) summarise the information on species traits of species recorded in both study sites, (ii) evaluate the relevance of individual traits for resisting drying out and (iii) verify the results of comparing the taxocenes of permanent and intermittent brooks.

The mayflies (Ephemeroptera) have been chosen as a suitable model group (while other benthic biota will be treated later), among others, the reasons are as follows. (i) First of all, in Central Europe with absolutely prevailing permanent water courses, data describing the effects of drought, on one hand and floods on the other (more precisely, unfavourable prediction scenarios not only in precipitation and runoff but also in temperature fluctuation) are still relatively very scarce or fragmentary; (ii) Mayflies, which first appeared in the fossil records of the Upper Carboniferous represent, with their only about 3,000 described species within 39 families a group in an evident regression with “old” and relatively firmly fixed adaptations, especially in larval stage; (iii) Due to a relatively very low vagility, their distribution is conservative with a very high degree of endemism (although not in the area studied and Europe in general); (iv) The Ephemeroptera order involves in general (as well as in taxocenes studied here) both less sensitive “generalists” with beneficial species traits as well as “specialists” with strict environmental limits and poor powers of dispersal that might be easily become extinct; (v) Mayflies have complex life cycles involving both aquatic and terrestrial phases. This type of life cycle creates evolutionary dichotomy with selection pressure operating in two, more or less, independent environments (cf., e.g., Wilbur 1980; or Brittain 1982, 1990, 1991 and others); (vi) In comparison with other benthic (and aquatic in general) groups of animals their extremely short-lived adult stages represent sole but crucial roles in reproduction, dispersal and recolonisation; (vii) Mayflies have survived, despite problems associated with selection processes operating in both aquatic and terrestrial environments, many climatic shifts and have successfully colonised a very wide range of freshwater habitats from the tropic to the Arctic and from small ponds to large rivers. For instance, in comparison with stoneflies (Plecoptera), they have made a greater intrusion into the tropics, both in terms of diversity and abundance, and are at the same time more abundant and diverse than dragonflies (Odonata) in the Arctic. This fact is extremely important because global climate change indications favour mayflies against Plecoptera and Odonata; (viii) Extremely low vagility of “conservative” mayflies with a very low presentation of purely behavioural adaptations most probably reflects the environmental changes much more sensitively than advanced and “progressive” insects groups such as, e.g.,

Table 1. Environmental parameters in the Gránický and Klaperův brooks.

	Gránický brook (intermittent)			Klaperův brook (permanent)		
	Min.	Max.	Average	Min.	Max.	Average
Water temperature (°C)	6.6	15.2	10.9	3.0	19.0	11.2
pH	6.1	8.6	7.8	6.3	7.9	7.3
Conductivity ($\mu\text{S cm}^{-1}$)	715	1090	884	594	692	643
Dissolved oxygen (mg L^{-1})	7.1	12.1	9.9	8.4	17.4	12.4
Depth (cm)	3	30	12	8	15	11

aquatic dipterans (Diptera), beetles (Coleoptera) and, in a lesser extent, also true aquatic bugs (Heteroptera–Nepomorpha) and caddisflies (Trichoptera) and, last but not least, (ix) mayfly species can be relatively easily identified in contrast with the above aquatic insect orders (perhaps except Heteroptera–Nepomorpha), because the larvae of numerous representatives do not provide us with sufficient critical diagnostic characters or even remain undescribed and (x) principal species traits of numerous mayfly species (or, more precisely, most species living in Central Europe) are relatively well known despite some gaps in knowledge, especially in rare species and often controversial literature data. This enables, in a relative easy way, to trace crucial species traits with inherent plasticity in response to environmental changes.

Study sites

Both the Klaperův and the Gránický brooks are third order tributaries of the Dyje (Thaya) River (the Danube catchment) situated in the warmest (mean annual air temperature is 8–9°C) and driest area (sum of annual precipitation is 450–500 mm) of the Czech Republic (Gránický brook: 48°51'59" N, 16°01'32" E; Klaperův brook: 48°52'24" N, 15°52'58" E) (Tolasz et al. 2007). The presence of sandy clay sediments and schistose biotitic granites with high concentrations of sulphates is the main reason for the higher values of conductivity of water typical for the region (Table 1).

The total length and catchment area of both streams is comparable (the Klaperův brook 8.5 km and 17.6 km²; the Gránický brook 13 km and 20.5 km², respectively). The brooks have similar substrate (diverse, but cobbles dominating; boulders are rare). The distance of both study sites, which are situated in the middle stretches of streams, is 10.5 km. The discharge of both brooks is comparable (ca. 0.2–0.3 m³ s⁻¹) during winter and spring. While the Klaperův brook has a permanent water discharge from source to mouth (except for the restricted uppermost segments), the upper part of the Gránický brook has a permanent character, and the middle part, where the study site is located, is intermittent. The lower part of brook is fed by groundwater and has a perennial character with drought events only in extremely dry years. The intermittent stretch became completely dry every summer for the past 12 years (1996–2008).

Material and methods

The data set

The basic data set proceeded from the period April 2002 – May 2003 when the benthic macroinvertebrates were sampled from both brooks at six-week intervals (except in winter). The data from the detailed study of the Gránický brook

(2005) were used to confirm the mayfly taxocene composition and study mayfly life cycles. Eleven series of samples were taken between April and November 2005 at three-week intervals. The semiquantitative multihabitat sampling method PERLA was used for both brooks and periods: multihabitat three-minute kick samples gathered with a hand net (25 × 25 cm aperture, mesh size 0.5 mm, sack length 75 cm) (CSN 75 7701; Kokeš et al. 2006). The material collected was fixed in 4% formaldehyde in the field, transferred into 75% alcohol after identification and deposited in collection at the Institute of Botany and Zoology, Faculty of Science, Masaryk University, Czech Republic. Physicochemical parameters (pH, conductivity, dissolved oxygen, water temperature; Table 1) and discharge were measured by portable instruments.

Species traits

Selected species traits of mayfly species found in the Gránický brook and/or in the Klaperův brook were compiled from original data and literature sources. Although species traits themselves have been treated and discussed not so frequently (e.g., Soldán & Zahrádková 2000; Brittain 2008; Zahrádková et al. 2009), appropriate (however, not complete) data concerning ecological requirements, dispersal and distribution of species in question are available in contemporary literature sources (see the list of references in Table 4). The most important seems to be tabular summaries of habitat/substrate preferences, feeding types, emergence, fecundity, oviposition and embryogenesis by Bauernfeind & Humpesch (2001), where numerous additional references can be found along with a biogeographical analysis by Haybach (1998, 2003, 2006) summarising also further references on the spatial and geographical occurrence and distributional types of Central European mayflies (Zahrádková et al. 2009). Although these data are controversial in some respects and some of them are still not known in details, we decided to avoid a detailed discussion in this respect (except the most important species trends concerning life cycles, see below). Moreover, some species traits (egg development and hatching, nymphal development, body size and shape and temperature relationships have been recently discussed by Brittain (2008) in detail with emphasis on their attributes generally advantageous or disadvantageous in disturbed habitats or in rapidly changing environments.

This paper focuses on the relationships between selected species traits on one hand and the effects of drought on the other in order to: (i) select critical species traits, (ii) define their polarity (i.e., which traits are advantageous and which are disadvantageous to survive drought) and (iii) define how much the individual species traits are pronounced within individual species found at the localities investigated.

The following traits were taken into account (see also Table 3 and 4): life cycle flexibility, bi- or polyvoltinism, adaptation in oviposition, egg quiescence, degree of parthenogenesis, asynchrony in emergence, length of winged

stage, larval body form, larval body length, feeding type, current velocity adaptation, habitat range, substrate range, density and distribution. The information about these traits was adopted from numerous sources (see Table 4); trait “density” is intended for mayfly taxocenes in the Czech Republic (Zahrádková et al. 2009).

An attribution of individual species traits as advantageous, disadvantageous or indifferent are apparent from Table 4. The attribution follows Brittain (2008) and general principles of r-K continuum concept (MacArthur & Wilson 1967; Pianka 1970).

To define the significance of individual species traits (i.e., how much are they pronounced in individual species), the three-grade scale was used to assess if the trait was advantageous for survival of drought events or not: 2 – advantageous; 1 – indifferent or less pronounced; 0 – disadvantageous or not pronounced. A total score was also calculated for each species in question (see Table 3 for details).

Life cycles represent the most important species traits concerning drought impacts. There are several systems of classification and nomenclature of mayfly life cycles. We used the classification by Clifford (1982), which seemed to be most frequently used in general (cf., e.g., Studemann et al. 1992; Haybach 1998, 2006; Sartori & Landolt 1998; Bauernfeind & Humpesch 2001; Derka 2003c). The Clifford classification evaluating the life cycle type as follows (only categories found in species dealt with in this study are mentioned). Main life cycle categories U – Seasonal univoltine; MB – Seasonal bivoltine, MP – Seasonal polyvoltine; U-MB – Total uni-multivoltine; Y – Total semivoltine; U-Y – Total uni-semivoltine. Life cycle groups (or “subcategories”): Uw – Seasonal univoltine (winter); Us – Seasonal univoltine (summer); Uw-Us – Seasonal univoltine (winter–summer); MBss – Seasonal bivoltine (summer); MBws – Seasonal bivoltine (summer–winter); MB-MP – Seasonal bivoltine or polyvoltine; Us-MBss – Seasonal variable (one or two summer generations); Uw-MBws – Seasonal variable (univoltine winter, possible summer generation); 2Y – Seasonal semivoltine (one generation in two years); 2Y-3Y – Seasonal semivoltine (one generation in one or two years); Uw-2Y – Seasonal variable (univoltine winter or two-year semivoltine)*.

* Because alternative classifications are often used (Landa 1968; Sowa 1975a), we consider it necessary to briefly present the equivalency among individual life cycle categories. In other words, to define which Clifford categories agree at least partially with those by Landa (1968) and/or Sowa (1975a). Life cycle subcategories according to the classification by Landa (1968) [main life cycle types A – Seasonal univoltine (winter or summer); B – Seasonal univoltine (winter–summer) and/or seasonal bivoltine and variable; C – Seasonal semivoltine (one generation in two or three years); D – Seasonal semivoltine (one generation in one or two years) and/or seasonal variable (univoltine winter or two-year semivoltine)] shows the following approximate relationships to subcategories by Clifford (1982): A1, A3 = Uw; A2 = Us; B1 = MBws; B2 = MBss and/or Us-MBss; B3 = MB-MP and/or Uw-MBws; B4 = Uw-Us; C1 = 2Y; C2 = 3Y; D1 = Uw-2Y; D2 = 2Y-3Y (here simplified, see Soldán & Zahrádková 2000 for further details). Main life cycle types according to the classification by Sowa (1975a) [main life cycle types: A – Seasonal semivoltine and/or seasonal variable (total uni-multivoltine), B – Seasonal univoltine (winter, summer, winter–summer); C – Seasonal bivoltine and/or polyvoltine and/or seasonal variable (one or two summer generations). Life cycle groups (subcategories) exhibit the following approximate relationships to subcategories by Clifford (1982): A1 = Uw-MBws and/or 2Y; A2 = Uw-2Y and/or 2Y-3Y; B1 = Us; B2 and/or B3 = Uw; B4 and/or B5 = Uw-Us; C1 = MBss; C2 = MBws; C3 = MB-MP (here simplified, see Soldán & Zahrádková 2000 for further details).

Results

Species composition of the mayfly taxocene

Altogether, 13 mayfly species of nine genera belonging to five families were found at the study sites (Table 2). Four species, namely *Siphonurus aestivalis* Eaton, 1903, *Baetis rhodani* (Pictet, 1843), *Electrogena* sp. cf. *ujhelyii* (Sowa, 1981) (see below) and *Habrophlebia fusca* (Curtis, 1834) formed the Ephemeroptera taxocene of the Gránický brook. Eleven species, namely *Baetis muticus* (L., 1758), *B. rhodani*, *Ecdyonurus starmachi* Sowa, 1971, *Rhithrogena carpa-toalpina* Klonowska, Olechowska, Sartori et Weichselbaumer, 1987, *Rhithrogena semicolorata* (Curtis, 1834), *Habroleptoides confusa* Sartori et Jacob, 1986, *Habrophlebia lauta* Eaton, 1884, *Paraleptophlebia submarginata* (Stephens, 1836), *Paraleptophlebia werneri* Ulmer, 1919 and *Ephemera danica* Müller, 1764 formed the mayfly taxocene of the Klaperův brook (see also Table 2 and 3).

Except for a single species of the genus *Electrogena*** of the family Heptageniidae (subfamily Heptageniinae), species collected at both sites of the area studied identified without any problems using current identification keys of Central European mayfly fauna (Landa 1969; Bauernfeind 1994, 1995; Bauernfeind & Humpesch 2001; Studemann et al. 1992).

Altogether 15 traits were estimated in all 13 species investigated, altogether 195 values presented (Table 3). The following species show (by the score) the highest ability to drought survival: *S. aestivalis* (24), *B. rhodani* (23), *B. muticus* (23), and *H. fusca* (18). Except for *B. muticus*, all these species were collected in the intermittent stream (Gránický brook). Further species collected in the intermittent stream, namely *E. sp. cf. ujhelyii* (12) seems to be much less adapted to surviving drought. The species found in the permanent Klaperův

** Concerning only the *Electrogena* species, there is a little doubt about its proper taxonomic position. Judging from the arrangement of larval morphological characters (mouthparts, leg and posterior margin of abdominal terga chaetotaxy, gills and cerci) and the exochorionic structure of eggs, this material most probably belongs to *Electrogena samalorum* (Landa, 1982), originally described sub *Ecdyonurus samalorum* Landa (Landa & Soldán 1982), also summarised critical distinguishing characters of this species from the closely related *Electrogena ujhelyii* (Sowa, 1981), originally described sub *Ecdyonurus ujhelyii* Sowa, 1981. Anyway, *E. samalorum* was found conspecific and synonymised with *E. ujhelyii* by Zurwerra & Tomka (1986), however, the type material had never been compared. Moreover, *E. ujhelyii*, originally misidentified as *Ecdyonurus subalpinus* Klapálek, 1907 (by Ujhelyi 1966, see Sowa 1981) had later been confused by the same authors (Tomka & Zurwerra 1985) with the related species *Electrogena gridellii* (Grandi, 1953) and *E. quadrilineata* (Landa, 1969) and described once again from Switzerland sub *Electrogena rivuscellana* Sartori & Landolt, 1991 (Landolt et al. 1991). The opinion of conspecificity of *E. ujhelyii* and *E. samalorum* is followed by some authors (e.g. Belfiore & Desio 1995; Bauernfeind & Humpesch 2001), whereas others (e.g., Derka 2003a, 2003b, 2003c in populations living in Slovakia) recognize the validity of *E. samalorum*. Because *E. samalorum* has still not been formally removed from synonymy with *E. ujhelyii* and its proper taxonomic position thus remains questionable, we present our material as *Electrogena sp. cf. ujhelyii*.

Table 2. Selected traits of mayfly species of the area studied.

Species	Site	Feeding group						Current preferences	Life cycle type at studied area	Alternative life cycle types
		Gránický Brook	Klaperův Brook	Grazers – Scrapers	Shredders	Gatherers – Collectors	Passive filter feeders			
<i>Siphonurus aestivalis</i> Eaton, 1903	*		1	+	9	0	0	RL	univoltine summer (Us-MBss)	Uw, Landa (1968): A2, B2
<i>Baetis muticus</i> (L., 1758)	*		5	0	5	0	0	RP	bivoltine winter-summer (MBws)	Uw, MP
<i>Baetis rhodani</i> (Pictet, 1843)	*		5	0	5	0	0	RP	bivoltine winter-summer (MBws)	Uw, MP
<i>Ecdyonurus starmachi</i> Sowa, 1971	*		7	0	3	0	0	RP	univoltine winter (Uw)	Sowa (1975b): B3
<i>Electrogena</i> sp. cf. <i>ujhelyii</i> (Sowa, 1981)	*		7	0	3	0	0	RP	univoltine winter (Uw)	MBws
<i>Rhithrogena carpatoalpina</i> Kłonowska, Olechowska, Sartori et Weichselbaumer, 1987	*		10	0	+	0	0	RB	univoltine winter (Uw)	
<i>Rhithrogena semicolorata</i> (Curtis, 1834)	*		10	0	+	0	0	RB	univoltine winter (Uw)	?MBws, ?Us-Uw, Landa (1968): A1, Sowa (1975a): B2
<i>Habroleptoides confusa</i> Sartori et Jacob, 1986	*		+	0	10	0	0	RP	univoltine winter (Uw)	Landa (1968): A1, Sowa (1975a): B2
<i>Habrophlebia fusca</i> (Curtis, 1834)	*		+	0	10	0	0	RL	univoltine winter (Uw)	Landa (1968): A3, Sowa (1975a): B1
<i>Habrophlebia lauta</i> Eaton, 1884	*		+	0	10	0	0	RL	univoltine winter (Uw)	?Us, Landa (1968): A3, Sowa (1975a): B3
<i>Paraleptophlebia submarginata</i> (Stephens, 1836)	*		+	0	10	0	0	RP	univoltine winter (Uw)	
<i>Paraleptophlebia werneri</i> Ulmer, 1919	*		+	0	10	0	0	RP	univoltine summer (Us)	Landa (1968): A2
<i>Ephemera danica</i> Müller, 1764	*		+	0	0	8	2	RP	semivoltine two years (2Y)	3Y, Uw, Landa (1968): C1, Sowa (1975a): A1

Explanations: Feeding types and current preferences according to Schmedtje & Colling (1996); RB (rheobiont), RP (rheophile), RL (rheo- to limnophile); for abbreviations of life cycle types see Material and methods.

brook only (except for *B. muticus* again) show generally lower scores (16 or less). The lowest adaptation to survive can be found, in this respect, in *R. semicolorata* (11), *E. starmachi* (9), and *R. carpatoalpina* (9).

Discussion

There is no doubt that the taxocene composition as well as survival and density of individual species found in the intermittent Gránický brook are influenced, first of all, by their life strategies and reproductive fitness. These attributes (now currently discussed as “species traits”) in fact represent a certain position of the species in r-K continuum. However, despite some exceptions (cf. Derka 2003c), r- and/or K-strategies have still not been discussed within the order Ephemeroptera, and the relative position(s) of species in question remains unknown in detail. Moreover, the situation seems to be rather complicated since, just in the Ephemeroptera, some apparent K-strategists may exhibit a “typical attribute” of r-strategy and *vice versa*. For instance, *E. danica*, a species

that might be considered a K-strategist with regards to some (prevailing) very conservative and fixed attributes (e.g., semivoltinism, no egg laying adaptations, no egg quiescence or survival of drought, larval body form and length, habitat, substrate range and others), showed extremely high fecundity (cf. Bauernfeind & Humpesch 2001), a very long winged stage and considerable flight ability – attributes considered “typical” for r-strategists. By contrast, *B. rhodani*, an apparent representative of r-strategists because of body form and length, bi- or polyvoltinism, life cycle flexibility, synchrony in egg hatching, larval development and emergence, pronounced adaptations in oviposition, density and distribution, showed very low fecundity (cf. Bauernfeind & Humpesch 2001), no respiratory and current velocity adaptation and a short winged stage life span – attributes considered “typical” for K-strategists.

Consequently, we avoid discussing the species found from this point of view because only very limited data concerning the other (here “reference” species) are available. Instead, we decided to evaluate species traits

Table 3. Selected species traits of mayfly species found in the (*) Klaperův brook and (**) Gránický brook, compiled from original data and literature sources.

Species found	<i>Siphonurus aestivalis</i>	<i>Baetis muticus</i>	<i>Baetis rhodani</i>	<i>Ecdyonurus starmachi</i>	<i>Electrogena</i> sp. cf. <i>ujhelyi</i>	<i>Rhithrogena carpatolpina</i>	<i>Rhithrogena semicolorata</i>	<i>Habroplectoides confusa</i>	<i>Habrophlebia fusca</i>	<i>Habrophlebia lauta</i>	<i>Paraleptophlebia submarginata</i>	<i>Paraleptophlebia werneri</i>	<i>Ephemera danica</i>
(*) – Klaperův Brook		(*)	(*)	(*)	(*)	(*)	(*)	(*)		(*)	(*)	(*)	(*)
(**) – Gránický Brook	(**)		(**)		(**)				(**)				
Selected traits and scores													
Life cycle flexibility	1	2	2	0	1	0	1	0	1	1	0	0	2
Bi- or polyvoltinism	1	2	2	0	0	0	1	0	0	0	0	0	0
Adaptation in oviposition	1	2	2	0	0	0	0	1	1	1	1	1	0
Egg quiescence	1	0	0	0	0	0	0	0	0	0	0	2	0
Degree of parthenogenesis	2	2	2	0	1	0	0	0	0	0	0	0	1
Asynchrony in emergence	1	2	2	1	2	1	1	1	1	2	1	0	2
Length of winged stage	2	0	0	2	2	2	2	2	1	1	2	1	2
Larval body form	2	2	2	0	0	0	0	1	1	1	1	1	0
Larval body length	1	2	2	1	1	1	1	1	2	2	1	2	0
Feeding type	2	1	1	1	1	0	0	2	2	2	2	2	0
Current velocity adaptation	2	1	1	1	1	0	0	1	2	2	1	1	1
Habitat range	2	1	1	0	0	0	0	1	2	1	1	2	1
Substrate range	2	2	2	1	1	1	1	2	2	1	2	1	1
Density	2	2	2	1	2	2	2	2	1	1	2	0	1
Distribution	2	2	2	1	0	2	2	2	2	1	2	0	2
Total score:	24	23	23	9	12	9	11	16	18	16	16	13	13

Explanations: Score: 0 – disadvantageous manifestation for drought survival; 1 – manifestation for drought survival not clearly pronounced or indifferent; 2 – advantageous manifestation for drought survival.

or, more precisely, selected traits apparently related to survival and (re-)colonisation.

There is a large number of species traits within mayflies, some of them clearly manifested and supported by the numerous data in the majority of Central European species (e.g., flexibility of life cycles, fecundity and length of emergence period), some of them described in detail only in some species (e.g., traits concerning the “ecological range”, see below) and others insufficiently known supported by concrete data only in a very small number of species (e.g., respiratory adaptation, true nature of quiescence or gene flow).

Furthermore, species traits can be manifested in different ways within different parts of species area. For instance, four types of developmental cycle have been identified in *B. rhodani*: (a) univoltine seasonal winter cycle (Uw) at latitudes above 65° and in mountains above 900–1,200 m a.s.l.; (b) bivoltine seasonal winter life cycle (MBws) or (c) seasonally variable cycle (Uw-MBws) mostly in a central latitudinal belt through Europe; and (d) seasonal polyvoltine cycle (MP) in southern area part, two summer generations have also been observed, for example in the Atlantic Pyrenees. Despite the occurrence of only 1–2 life cycle types within Central Europe in general, (b) and/or (c) (cf. Landa 1968; Sowa 1975a; Studemann et al. 1992; Sartori & Landolt 1998; Bauernfeind & Humpesch 2001), this species

shows the highest life cycle flexibility of all remaining species at both localities.

Finally, much literature data seem to be highly controversial. This concerns for example the data on adaptations for oviposition or actual fecundity in *S. aestivalis*. In the latter case, there are differences from several hundreds to 2–3 thousand eggs per female and fecundity largely depends on body size, generation, season and different local conditions (see Soldán & Zahradková 2000; Bauernfeind & Humpesch 2001 for a complete list of respective references). Nevertheless, *S. aestivalis* shows relatively high fecundity, however, on our scale it was comparable to most species of the Heptageniidae. This was the reason we avoided to consider fecundity in our localities studied, however, it could represent very important species trait in general.

Naturally, the significance of individual species traits is rather different. Besides so-called “ecological species traits”, e.g., habitat and substrate range, density, distribution, and current velocity adaptation, we consider so-called “biological species traits” more important to explain survival and recolonisation (see below). Generally, there are three main types of biological traits: (i) those connected with life cycle are the most important and probably determine survival from generation to generation. Besides the type of life cycle (see Material and methods for their delimitation) this

Table 4. Nature (polarity) of species traits with regards to drought survival in species investigated (see Table 3 for the species list), compiled from original data and literature sources.

Selected species trait	Advantageous manifestation for drought survival (score: 2)	Manifestation for drought survival not clearly pronounced or indifferent (score: 1)	Disadvantageous manifestation for drought survival (score: 0)
Life cycle flexibility (2, 3, 5, 12, 18, 19)	flexible (regularly two or several life cycle types)	not strictly fixed (usually a single life cycle type, at most one alternative life cycle type)	strictly fixed (the only life cycle type)
Bi- or polyvoltinism (2, 3, 4, 5, 12, 18, 19)	present (regularly more than 1 gen./year)	possible (usually univoltine with possible complete or incomplete 2nd generation)	absent (regularly a single gen./year or semivoltinism)
Adaptation in oviposition (1)	female underwater, oviposition on substrate, ovoviviparity	female above water, oviposition on substrate	female flying, toughing water surface
Egg quiescence (3, 4, 5, 7, 12, 18)	long (always overwintering eggs, mostly with diapausis)	variable (eggs may overwinter or not, mostly without diapausis)	short (eggs never overwinter, diapausis absent)
Degree of parthenogenesis (6, 7, 13, 15)	high (>30 %)	medium (5–30 %)	negligible (<5 %)
Asynchrony in emergence (1, 8, 9)	high (eggs hatch regularly in several cohorts)	medium (eggs hatch usually in a single cohort)	low (eggs hatch regularly in a single cohort)
Length of winged stage life span (13, 15)	long (more than 2 days)	medium (1–2 days)	short (1 day or shorter)
Larval body form (4, 8, 13)	not specialised, cylindrical (fish-like)	partly specialised (sprawling, walking)	specialised – adapted (flattened, hydrodynamic, burrowing)
Larval body length (4, 9, 13)	small (≤ 10 mm)	medium (10–20 mm)	large (>20 mm)
Feeding type (8, 16, 19)	gatherers-collectors, omnivores	mixed types (gatherers-collectors and grazers-scrappers)	filter feeders or grazers-scrappers specialists
Current velocity adaptation (8, 16, 19)	rheo- to limnophil	rheophile	rheobiont
Habitat range (1, 8, 14, 16, 19)	broad (colonizing more 5 or more habitat)	medium (colonizing 3 or 4 habitats)	narrow (colonizing usually a 1 or 2 habitat)
Substrate range (16, 19)	broad (living in substrates with different roughness)	medium (living in substrates with similar roughness)	narrow (specialized, living in substrates with defined roughness)
Density (16, 19)	high (eudominant or dominant species)	medium (subdominant or recedent species)	low (subrecedent species)
Distribution (1, 8, 9, 10, 11, 16, 17, 19)	large areas (Palaeartic), very frequent or frequent localities in Central Europe	medium areas (West Palaeartic), medium frequent or scarce localities in Central Europe	small areas (submediterranean), very scarce localities in Central Europe

Explanations: 1 – Bauernfeind & Humpesch (2001), 2 – Brittain (1990), 3 – Brittain (1991), 4 – Brittain (2008), 5 – Clifford (1982), 6 – Degrange (1954), 7 – Degrange (1960), 8 – Derka (2003c), 9 – Haybach (1998), 10 – Haybach (2006), 11 – Landa & Soldán (1985), 12 – Landa (1968), 13 – Landa (1969), 14 – Sartori & Landolt (1999), 15 – Soldán, unpubl., 16 – Soldán & Zahrádková (2000), 17 – Sowa (1975a), 18 – Sowa (1975b), 19 – Zahrádková et al. (2009)

category involves also tendency to polyvoltinism, flexibility of the cycle, length of embryogenesis, egg quiescence, degree of parthenogenesis, asynchrony in egg hatch, development and emergence; (ii) species traits connected with the adult stage, particularly important as far as the vagility of mayflies is concerned (e.g., egg laying adaptation and length of winged stage); and (iii) species traits connected with the larval stage and basic physiological functions (e.g., larval body form and length).

Intermittent and permanent brook: comparison of mayfly taxocene survival and recolonisation

We compared our study site with the adjacent Klaperův

brook to evaluate changes in taxocene composition under unstable hydrological regime contra expected (reference) status of permanent brook of the same type. Judging from respective chemical analyses, the conspicuous impoverishment of taxocene components is definitely not caused by the pollution of water and thereby most likely represents a consequence of an anomalous hydrological regime, a key factor governing the mayfly taxocene composition at least in this case. Moreover, the total absence of pronounced rheobiont species indicates that a very important role is played by current velocity.

The differences in survival in the intermittent Gránický brook and adjacent permanent Klaperův

brook, which are probably closely related in colonisation and recolonisation cycles, are expressed by the occurrence of only four species in the former while the latter permanent brook is inhabited by additional nine species (Tables 2 and 3), e.g., by rheobionts *R. semicolorata* and *R. carpatoalpina* or high oxygen demand species such as *E. starmachi*, or semivoltine non-seasonal species showing very plastic developmental cycle such as *E. danica* occurring in relatively high densities. The latter species, although exhibiting a relatively high total score (Table 3) enabling larvae to occur and easily survive at high densities in the Klaperův brook, is not able to persist in an intermittent brook. Most probably, larval survival is prevented by their relatively high oxygen demands. Contrary to some other species of *Ephemera*, larvae of *E. danica* never occur in lakes, ponds and other standing waters. In addition, traits such as substrate range, body form, length and especially apparent semivoltinism with no egg dormancy are apparently unfavourable (cf. Brittain 2008).

Total scores (Table 3), clearly show the species with the highest number of advantageous species traits, in other words, the most successful species for survival and recolonisation under the unfavourable conditions of the intermittent brook, namely *S. aestivalis* and *B. rhodani*. However, both species seem to utilise different strategies to survive. Survival of the former is enabled mainly by a large ecological range of larvae with apparently low oxygen demands and respiratory adaptation (movable gills). Larvae are also known to survive in periodic water bodies and usually quickly develop in two months in spring during relatively good water supply. Adults fly in May and July. The eggs (hatching usually early next spring with a high degree of parthenogenesis; cf. Degrange 1954, 1960) are able to tolerate drying up during summer months (cf. Bohle & Potabgy 1992 or Fiedler & Bohle 1994). Sættem & Brittain (1993) even pointed out the summer diapause (aestivation) in populations living in relatively very warm water. *S. aestivalis* was not found in the Klaperův brook at the places sampled during this study but it occurs, however in very small abundance, at several localities situated upstream (in permanent brook segments that might be subjected to severe drought) (Nyklová 2006).

By contrast, *B. rhodani* seems to be short of the pronounced attributes enabling survival of *S. aestivalis* because the larvae are apparently adapted to stream-line habitats. Survival is enabled by oviposition adaptation and mainly enormous flexibility of life cycle accompanied with asynchrony in embryogenesis, larval growth and emergence. There is no doubt that at least some larval cohorts are able to survive partial drying up or recolonise brook segments that have been subjected to total drought. The taxocene of the intermittent brook is short of *B. muticus*, although this species exhibits a relatively high number of advantageous species traits. The absence of the species can be explained by (i) a competition with *B. rhodani* – both species have similar ecological requirements, but *B. rhodani* is less specialised (Schmedtje & Colling 1996; Zahrádková et al.

2009) and thus probably more successful in survival; and (ii) *B. muticus*, in spite of relatively high number of “advantageous” traits for drought survival, is considered a species sensitive to anomalous hydrological regimes (Céréghino et al. 2002). This fully agrees with our opinion that *B. muticus* belongs to species first disappearing in intermittent brooks or running waters subjected to non-periodical water level fluctuation, as seen at numerous localities in the same area (Zahrádková unpublished).

As to *E. sp. cf. ujhelyii*, its life cycle and details on species traits are far from to be understood, published data are extremely scarce. However, the total score of species traits in *E. sp. cf. ujhelyii* is relatively low in comparison with survival of other species, it is the highest within the Heptageniidae studied that represent the most sensitive species in general. Larvae with respiratory adaptation (movable gills of most pairs) living among submerged roots or logs at rather slow current places are probably able to survive partial drought. Moreover, they apparently develop in several cohorts and adults show a pronounced asynchrony in emergence, they fly from April to September (cf. e.g., Bauernfeind & Humpesch 2001).

The occurrence and survival of *H. fusca* (living in the Gránický brook) and *H. lauta* (living in the Klaperův brook) agree with the differences in our total scores (Table 3) as well as with literature data. Larvae of *H. fusca* are evidently much better adapted to survive unfavourable conditions, including at least partial drought as documented, e.g., in Germany (Fiedler & Bohle 1994), Spain (Gallardo-Mayenco & Ferreras Romero 1984) and North Africa (El Agbani et al. 1992). Contrary to *H. lauta*, the species is considered azonal and thermophilous (Haybach 1998; Jacob 1972), generally tolerating large substrate and habitat range (Landa 1957, 1969; Sowa 1975a; Sartori & Landolt 1999; Derka 2003b).

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